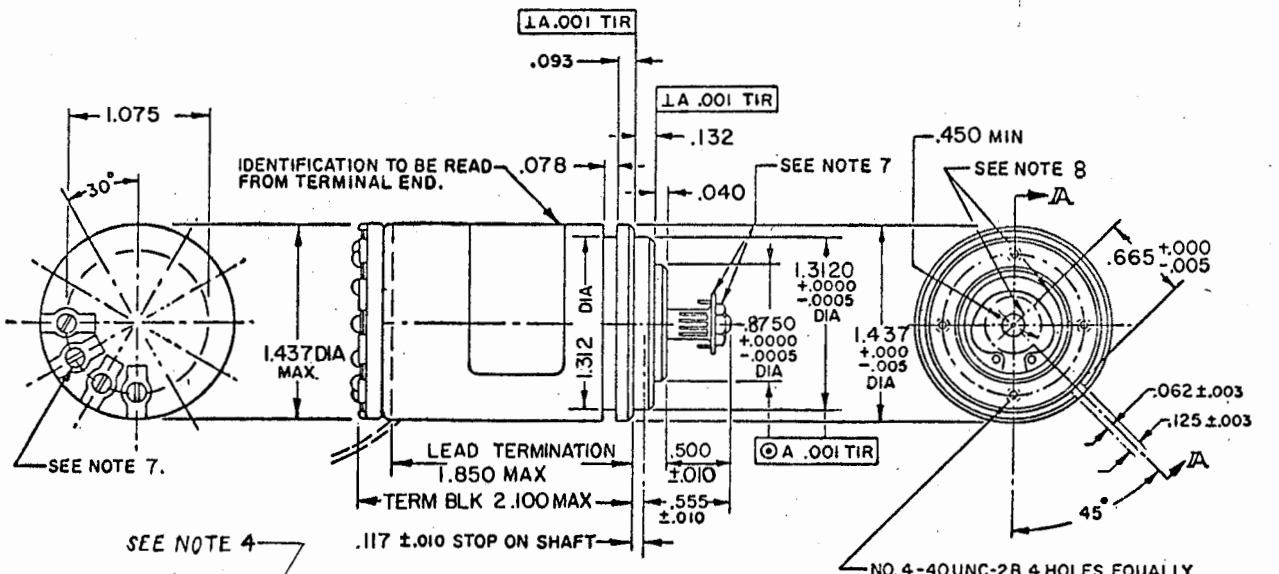
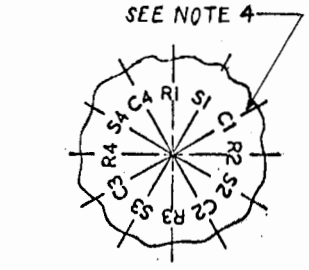
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL

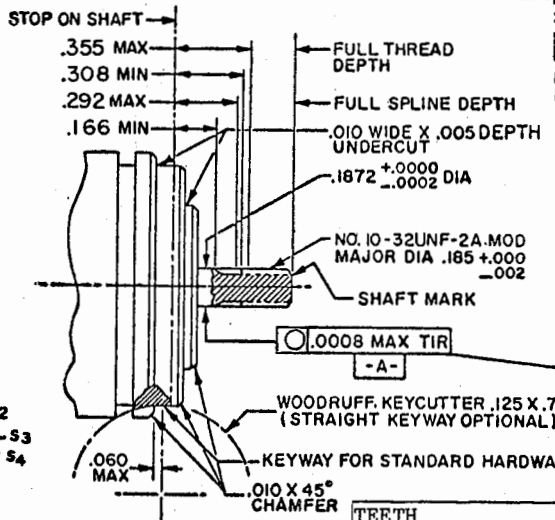
TYPE 15R11W4a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.

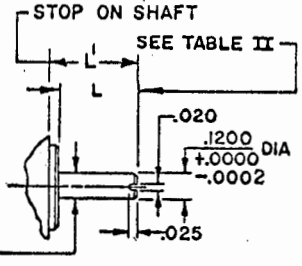




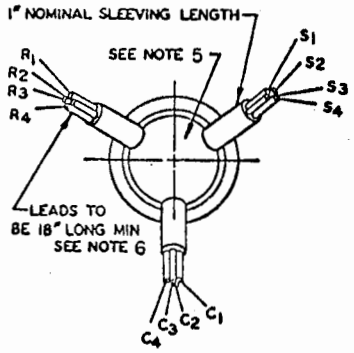
END VIEW- ENLARGED



PARTIAL SECTION A-A



PLAIN AND SLOTTED SHAFT



LEAD TERMINATION

SPLINE DATA

TEETH	21
PITCH	120
PITCH DIAMETER	.175+.000-.002
OUTSIDE DIAMETER	.1872+.0000-.0005
PRESSURE ANGLE	20°
ROOT DIA. MAX	.155
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE MS90406.

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. Mounting screw positions optional. Terminal block designed for terminal lug.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Leads shall conform to MIL-W-16878/4, type E-28.
7. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatable with the weight of the contents. The packaged hardware shall be placed in the unit resolver package as follows:

Machine screw	(MS35275-213)	12 required (*) (**)
Lockwasher	(MS35338-135)	12 required (*) (**)
Drive washer	(MS17186-6)	1 required (*)
Nut	(MS17187-2)	1 required (*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.
For information only: Use mounting clamp MS17183 or equal.

8. Mark on housing is to match shaft mark within 10° when rotor is set at approx. resolver zero, and will bear no relation to the center lines shown on the drawing.
9. A voltage value of 5 to 26 V shall be entered in the 'Volts' designation space on the identification plate.
10. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.
11. With the resolver energized under standard test conditions the resonant frequency will be recorded during qualification as that point where the output voltage has a maximum value and phase shift -90°. Low corner frequency (referenced as design information) is the point where output voltage is down 3db and phase shift 45°.
12. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.

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TABLE I

RESOLVER, ELECTRICAL: TYPE 15R11W4a

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

	Requirement	Value	Unit	Tolerance
1.	Excitation winding	Stator		
2.	Rated frequency	400	Hz	±1%
3.	Input voltage		volts (rms)	±1%
	Maximum	26		
	Test	15		
	Minimum	5		
4.	Primary current	14	milliamps	nominal
5.	Primary power	0.044	watts	nominal
6.	D. C. resistance		ohms	±15%
	Rotor	130		
	Stator	165		
	Compensator	215		
7.	Impedance		ohms	R±15%+jx±10%
	Z_{RO}	200+j1200		
	Z_{SO}	225+j1050		
	Z_{CO}	275+j1100		
	Z_{RS}	285+j125		nominal
8.	Transformation ratio			
	Output/input	0.98		±1%
	Rotor/compensator	0.98		±0.1%
9.	T. R. variation (R/C)		%	
	Due to voltage	0.05		maximum
10.	Equality of transformation ratios		% spread	maximum
	Output/input	0.1		
	Rotor/compensator	0.1		
11.	Phase shift		degrees	
	Output/input	8.5		±0.5
	Rotor/compensator	8.5		±0.5
12.	Phase shift variation (R/C)		degrees	
	Due to voltage	±0.5		
13.	Interaxis error	± 5	minutes	
14.	Functional error	±0.1	%	
15.	Null voltage		mV(rms) output	maximum
	Fundamental	8		
	Total	15		
16.	Shift of resolver zero		minutes	
	With voltage	±1.0		
	With frequency	± .05		
17.	Harmonic distortion	0.1	%	maximum

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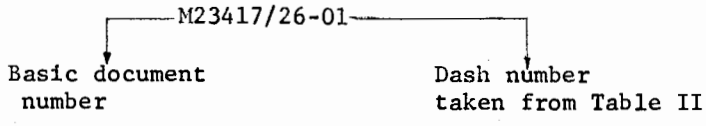
TABLE I (Continued)

	Requirement	Value	Unit	Tolerance
18.	Temperature rise	15	°C	maximum
19.	Friction torque	0.15	oz. - in.	maximum
	@ -55°C	0.30		maximum
20.	Radial play	0.0005 (for reversal of 1/2 lb load)	inches	maximum
21.	End play	0.0010 (for reversal of 1 lb load)	inches	±0.0005
22.	Wirelead stress (see note 10)	2	lbs	minimum
23.	Resonant frequency (see note 11)		kHz	minimum
24.	Low corner frequency (see note 11)	63.5	Hz	maximum

NOTE: This unit is a preferred version of the MK4 Mod 0 and is recommended for new design. The MK4 Mod 0 is covered by MIL-R-23417/6(WP).

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from Table II.

Example:



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TABLE II

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				Free length - L (±.010)	Dim to stop on shaft - L' (±.010)
-01	.187	Spline	Terminal block	.500(1/2)	.555
-02	.120	Plain	Wire leads	.250(1/4)	.305
-03	.120	Plain	Wire leads	.437(7/16)	.492
-04	.120	Plain	Wire leads	.500(1/2)	.555
-05	.120	Plain	Terminal block	.250(1/4)	.305
-06	.120	Plain	Terminal block	.437(7/16)	.492
-07	.120	Plain	Terminal block	.500(1/2)	.555

Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - MU, EL
Navy - AS, SH, OS
Air Force - 11, 17, 85

User activities:
Army - MI, AV, ME, AT
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - MU

(Project 5990-0111)

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